

Problems

Problem 1 (12 pts., 2 pts. each): Condition Strength

Order conditions from **the strongest to the weakest** by showing the longest implication chain of the form $A \rightarrow B \rightarrow C \rightarrow D \dots$ (where $A, B, C, D, \dots$ are given conditions, $\rightarrow$ is implication). If total ordering by strength cannot be established, show all partial orderings. Prove by showing all implications. If any implication is false, provide a counterexample. You must show all work in order to get full credit. Unless otherwise stated, assume all referenced variables are defined as integers.

- (1) $A = \{ x = 2k + 1 \wedge x = x + 1 \wedge y = 10 \}$
 $B = \{ x \text{ is divisible by } 6 \wedge y = x + 2 \}$
 $C = \{ x = 18 \wedge y = 20 \}$
 $D = \{ x \% 2 = 0 \wedge y \text{ is even} \}$

Implication Chain: $() \rightarrow () \rightarrow () \rightarrow ()$

Proof of implication: $A \rightarrow B : \dots$

$A \rightarrow B$: We know A is false simply because $x = x + 1$ cannot be true as no integer fits this condition. Thus, A will always be false and implies other conditions making it true. Conversely, $B \rightarrow A$ is false as a truth implying a falsehood results in a falsehood.

$B \rightarrow C$: This implication does not work. For B , we could say $x = 12$ and $y = 14$. Now let's try $C \rightarrow B$. When $x = 18$ and $y = 20$, we can say for certain that x is divisible by 6 and $y = x + 2$.

$B \rightarrow D$: All numbers divisible by 6 are even, and all multiples of 6 are even. Adding 2 to these multiples still keeps them even. Thus, the implications holds. Conversely, $D \rightarrow B$ does not hold. In this case, we could say that $x = 12$ and $y = 14$, which would make C false.

Thus, our implication chain is: $A \rightarrow C \rightarrow B \rightarrow D$

- (2) $A = \{ 5 \leq k < 5 \}$
 $B = \{ k \geq -10 \}$
 $C = \{ -10 < k \leq 1 \}$
 $D = \{ 10 \leq k \leq -10 \}$

Implication Chain: $() \rightarrow () \rightarrow () \rightarrow ()$

Proof of implication: $A \rightarrow B : \dots$

$A \rightarrow B$: A can never be true, and since B can be true, the implication is always false. Conversely, $B \rightarrow A$ as an implication can be true.

$A \rightarrow C$: Since A can never be true, and C can be, the implication is always false. Conversely, $C \rightarrow A$ as an implication can be true.

$A \rightarrow D$: Both A and D can never be true. Since two falsehoods making a truth is the strongest implication, we will use this going forward. This argument can be made both directions since both are false.

$D \rightarrow C$: This implication is strongest as while both B and C are true, B is less specific. Thus, our stronger implication is $D \rightarrow C$.

$C \rightarrow B$: Both these conditions can be true. As it stands, any number for k would work in this implication. Conversely, if we had $B \rightarrow C$, we could say $k = 100$, which would not come out true for C , making the implication false.

Thus, our implication chain is: $A \rightarrow D \rightarrow C \rightarrow B$

- (3) $A = \{ x \geq 0 \wedge y > x \}$
 $B = \{ x = 3 \wedge y > 10 \}$
 $C = \{ x = y \% 10 \}$
 $D = \{ xy = 0 \}$
 $E = \{ x > 1 \vee y > 1 \}$

Implication Chain: $() \rightarrow () \rightarrow () \rightarrow () \rightarrow ()$
 Proof of implication: $A \rightarrow B : \dots$

$A \rightarrow B$: Counterexample: $x=4$ and $y=5$
 $B \rightarrow A$: Holds true under all conditions
 $A \rightarrow C$: Counter example: $x=3$ and $y=11$
 $A \rightarrow D$: Counterexample: $x=3$ and $y=11$
 $A \rightarrow E$: Holds true under all conditions
 Partial ordering: $B \rightarrow A \rightarrow E$

- (4) Recall the names for infinite sets of numbers used in math (e.g., see “Peter Jephson Cameron (1998). *Introduction to Algebra*. Oxford University Press. p. 4. ISBN 978-0-19-850195-4.”)

$$A = \{ z \in \mathbb{Q} \}$$

$$B = \{ z = \sqrt{-1} \}$$

$$C = \{ z \in \mathbb{N} \}$$

$$D = \{ z = m / n \}$$

$$E = \{ z \in \mathbb{Z} \}$$

$$F = \{ z = y \% 2 \}$$

$$G = \{ z \in \mathbb{R} \}$$

$$H = \{ z = y / 2 \}$$

Implication Chain: $() \rightarrow () \rightarrow () \rightarrow () \rightarrow () \rightarrow () \rightarrow () \rightarrow ()$

Proof of implication: $A \rightarrow B : \dots$

$$F \rightarrow C \rightarrow E \rightarrow A \rightarrow G$$

This partial ordering works as we can start with any number for y, which results in z being either 1 or 0. Both these numbers are integers, which are natural numbers, which are integers, which are rational numbers, which are real numbers.

- (5) $A = \{ -7 < x < 0 \}$

$$B = \{ y = \log(x) \}$$

$$C = \{ -5 < x \leq 10 \}$$

$$D = \{ x = 1 \wedge y \geq 1 \}$$

$$E = \{ -1 \leq x \leq 1 \}$$

Implication Chain: $() \rightarrow () \rightarrow () \rightarrow () \rightarrow ()$

Proof of implication: $A \rightarrow B : \dots$

$$D \rightarrow E$$

This ordering works as since x will always be 1, and y will be greater than 1, x fits the conditions of E by falling between 1 and -1 inclusive.

$$D \rightarrow C$$

This ordering works for the same reasons as before, with x always being between -5 and 10. Both orderings cannot be combined as the implications between C and E would violate each other in either direction.

- (6) Assume "result is a double" and $|\cdot|$ is the absolute value operator.

$$A = \{ |\text{result} - \sin(x)| \leq 1 \}$$

$$B = \{ |\text{result} - \sin(x)| \leq 0.01 \}$$

$$C = \{ |\text{result} - \sin(x)| \leq 10^{-10} \}$$

$$D = \{ |\text{result} - \sin(x)| \leq -0.01 \} = \{ \text{false} \}$$

$$E = \{ |\text{result} - \sin(x)| \geq x^2 \}$$

Implication Chain: $() \rightarrow () \rightarrow () \rightarrow () \rightarrow ()$

Proof of implication: $A \rightarrow B : \dots$

All conditions here can be disproven by some result that is a double, as doubles can be negative and positive and have up to 15 digits, and be decimals.

Problem 2 (4 pts., 1 pt. each): Hoare Triples

State whether each Hoare triple is valid. If it is invalid, explain why and show how you would modify the the postcondition to make it valid. Unless otherwise stated, assume all referenced variables are defined as integers.

(1) $\{ x = 5 \}$

$$x = x * 2;$$

$$\{ x = 10 \vee x \neq 0 \}$$

Valid or Invalid: (...)

When $x = 5$, the statement makes $x = 10$, and thus the postcondition is correct. Hoare Triple is valid.

(2) $\{ \sqrt{x-1} > k \}$

$$x = x + 1;$$

$$\{ k \geq 0 \}$$

Valid or Invalid: (...)

If we assume $x = 2$, and the precondition is true, then the statement results in $x = 2$. If we are to follow the logic, k could be anything less than 2. Therefore, the hoare triple is invalid. To make this hoare triple valid, we can change the postcondition to: $\{ x > k \}$.

(3) $\{ i + j \neq 0 \wedge i \cdot j \neq 0 \}$

$$i = j - 1;$$

$$j = i + 1;$$

$$\{ (i = 0 \vee i \neq -j) \wedge k \in \mathbb{Q} \}$$

Valid or Invalid: (...)

To prove this hoare triple, we must assume that both integers i and j are either positive or negative. If one is negative and the other is positive, we could have a situation where they equal zero. In addition, we must assume neither is equal to zero, so that when multiplied the product does not result in zero. Now, lets follow the logic when $i = 100$ and $j = 2$. Following the first statement, $i = 1$. Following the second statement, $j = 2$. In any scenario, j will remain the same, and i will end up being a number one less than j . Now, lets look at the postcondition. Either, $i = 0$ or i cannot equal $-j$. The second of these is satisfied. Then, we must be able to say that k is a rational number. This hoare triple is invalid because we have no way of knowing what k is, and therefore it could be anything. To make this triple valid, we would change the postcondition to: $\{ i = 0 \vee i \neq -j \}$

(4) $\{ n < 0 \wedge n = \sqrt{m} \}$

```
if (n > m)
  x = n;
else
  y = m;
```

{ $x \neq y$ }

Valid or Invalid: (...)

To prove this hoare triple true, we must follow the preconditions. Let's assume $n = -10$ and $m = 100$. Now our postconditions are satisfied. Following the statements, n can never be greater than m , so we would always move to the else statement. we can now state that $y = 100$. Our postcondition, however, makes the statement invalid, as we never will find out what x is. Thus, this hoare triple is invalid. To make the hoare triple valid, we will change the postcondition to: { $x \neq m \vee y \neq 0$ }

Problem 3 (4 pts., 1 pt. each): General Hoare Triples

A, B, C, D, E, and F are logical conditions (logical formulas). The following are true:

- $A \rightarrow B$ (A implies B, i.e., A is stronger than B)
- $B \rightarrow C$
- $C \rightarrow B$
- $D \rightarrow E$
- $E \rightarrow F$
- $\{ B \} \text{ code } \{ E \}$

Indicate whether the following Hoare triples are valid or possibly invalid. If possibly invalid, prove by giving an example.

(1) $\{C\} \text{ code } \{D\}$ Valid or Possibly Invalid: (...)

If we have $A \wedge B \wedge C$, $B \text{ code } E$, $D \wedge E \wedge F$

This triple is possibly invalid.

Example:

B: $x > 0$

C: $x > 0$

D: $x = \text{All real numbers}$

E: $x > 10$

code: $x = x + 5$

If we say $x = 1$ for B, and $x = 6$ from the code, which does not satisfy E.

(2) $\{B\} \text{ code } \{C\}$ Valid or Possibly Invalid: (...)

This triple is valid.

(3) $\{A\} \text{ code } \{D\}$ Valid or Possibly Invalid: (...)

This triple is possibly invalid.

Example:

A: $x > 5$

B: $x > 0$

D: $x > 30$

E: $x > 12$

code: $x = x + 5$

If we say $x = 6$ for B, and $x = 11$ from the code, which does not satisfy E.

(4) $\{A\} \text{ code } \{F\}$ Valid or Possibly Invalid: (...)

This triple is possibly invalid.

Example:

A: $x > 10$

B: $x > 0$

D: $x > 100$

E: $x > 30$

F: $x > 20$

code: $x = x + 5$

If we say $x = 11$ for A, and $x = 16$ from the code, which does not satisfy F.

Problem 4 (8 pts., 1 pt. for each condition): Forward reasoning

For each code snippet with the given precondition, find the **strongest postcondition** by inserting the appropriate condition in each blank. The first intermediate condition in part (1) is supplied as an example. Please simplify your answers as much as possible. Assume all referenced variables are defined as integers.

Copy all code to your answer file and fill in the blanks. Carry all variables forward. Show all work.

(1) { $z \neq 0$ }
 $y = 0$;
 { $y = 0 \wedge z \neq 0$ }
 $x = y + 2$;
 { $y = 0 \wedge z \neq 0 \wedge x = 2$ }
 $z = x + y$;
 { $y = 0 \wedge z = 2 \wedge x = 2$ }

- (2) $\{ |x| > 5 \}$
 $x = x \% 10;$
 $\{ 0 \leq x \leq 9 \}$
 $x = x * x;$
 $\{ 0 \leq x \leq 81 \}$
 $x = -x;$
 $\{ -81 \leq x \leq 0 \}$
- (3) $\{ z < 5 \}$
 $\text{if } (z \geq 0) \{$
 $\{ 0 \leq z < 5 \}$
 $z = -z;$
 $\{ -5 < z \leq 0 \}$
 $\}$
 $\{ -5 < z \leq 0 \vee z < 5 \}$

Problem 5 (11 pts., 0.5 pts. each condition): Backward reasoning

Find the **weakest precondition** of each code sequence by inserting the appropriate condition in each blank. The first intermediate condition in part (1) is supplied as an example. Please simplify your answers as much as possible. Assume all referenced variables are defined as integers. Use the $\text{wp}()$ notation as shown in the first example.

- (1) $\{ \text{wp}("x = -1; z = 3 * x + y; ", 0 < z < 10) \} = \{ z = -3 + y \wedge 0 < z < 10 \} =$
 $\{ 0 < -3 + y < 10 \} = \{ 3 < y < 13 \}$
 $x = -1;$
 $\{ \text{wp}("z = 3 * y + x; ", 0 < z < 10) \} = \{ 0 < 3x + y < 10 \} =$
 $\{ y > -3x \wedge y < 10 - 3x \}$
 $z = 3 * x + y;$
 $\{ 0 < z < 10 \}$
- (2) $\{ \text{wp}("y > 0 \vee (y \leq 0 \wedge x > 1); ", x > 1 \wedge y = 0) \} = \{ (x > 1 \wedge y = 0) \}$
 $\text{if } (y > 0) \{$
 $\{ \text{wp}("x = x / y \wedge y > 0; ", x > 1 \wedge y = 0) \} = \{ x > 1 \wedge y = 0 \}$
 $x = x / y;$
 $\{ \text{wp}("y = 0; ", x > 1 \wedge y = 0) \} = \{ x > 1 \wedge y = 0 \}$
 $y = 0;$
 $\{ \text{wp}(";", x > 1 \wedge y = 0) \} = \{ x > 1 \wedge y = 0 \}$
 $\}$ $\text{else } \{$
 $\{ \text{wp}("y = 4 * x; ", x > 1 \wedge y = 0) \} = \{ x > 1 \wedge y > 4 \}$
 $y = 4 * x;$
 $\{ \text{wp}(";", x > 1 \wedge y = 0) \} = \{ x > 1 \wedge y = 0 \}$
 $\}$
 $\{ x > 1 \wedge y = 0 \}$

- (3) $\{ \text{wp}("x \geq 0 \vee x < 0;", z \neq 0 \wedge y \geq 0 \vee x \geq 0) \} = \{ (z \neq 0 \wedge y \geq 0 \wedge x < 0) \vee x \geq 0 \}$
 if $(x \geq 0)$ {
 $\{ \text{wp}("x \geq 0 \wedge z = \text{Math.min}(z, x);", z \neq 0 \wedge y \geq 0 \vee x \geq 0) \} =$
 $\{ x \geq 0 \wedge z = \text{Math.min}(z, x) \}$
 $z = \text{Math.min}(z, x);$
 $\{ \text{wp}("x = z + y;", z \neq 0 \wedge y \geq 0 \vee x \geq 0) \} =$
 $\{ z \neq 0 \wedge y \geq 0 \vee x \geq 0 \}$
 $x = z + y;$
 $\{ \text{wp}(";", z \neq 0 \wedge y \geq 0 \vee x \geq 0) \} = \{ z \neq 0 \wedge y \geq 0 \vee x \geq 0 \}$
 }
 $\{ z \neq 0 \wedge y \geq 0 \vee x \geq 0 \}$
- (4) $\{ \text{wp}(" \text{Math.abs}(x) \leq 5 \vee \text{Math.abs}(x) > 5;", -3 \leq z \leq 3) \} =$
 $\{ (\text{Math.abs}(x) \leq 5 \vee \text{Math.abs}(x) > 5) \}$
 if $(\text{Math.abs}(x) \leq 5)$ {
 $\{ \text{wp}("z = x - 2;", -3 \leq z \leq 3) \} = \{ -3 \leq x - 2 \leq 3 \} =$
 $\{ -1 \leq x \leq 5 \}$
 $z = x - 2;$
 $\{ \text{wp}(";", -3 \leq z \leq 3) \} = \{ -3 \leq z \leq 3 \}$
 } else {
 $\{ \text{wp}("x \leq -5 \vee x > -5;", -3 \leq z \leq 3) \} = \{ x \leq -5 \vee x > -5 \}$
 if $(x \leq -5)$ {
 $\{ \text{wp}("z = x + 6;", -3 \leq z \leq 3) \} = \{ -3 \leq x + 6 \leq 3 \} =$
 $\{ -9 \leq x \leq -3 \}$
 $z = x + 6;$
 $\{ \text{wp}(";", -3 \leq z \leq 3) \} = \{ -3 \leq z \leq 3 \}$
 } else {
 $\{ \text{wp}("z = 3 * x;", -3 \leq z \leq 3) \} = \{ -3 \leq 3x \leq 3 \} =$
 $\{ -1 \leq x \leq 1 \}$
 $z = 3 * x;$
 $\{ \text{wp}(";", -3 \leq z \leq 3) \} = \{ -3 \leq z \leq 3 \}$
 }
 $\{ \text{wp}(";", -3 \leq z \leq 3) \} = \{ -3 \leq z \leq 3 \}$
 }
 $\{ -3 \leq z \leq 3 \}$
- (5) $\{ \text{wp}("x = y / 2;", x \neq 0.5 \wedge z > 0) \} = \{ x \neq 0.5 \wedge z > 0 \}$
 $x = y / 2;$
 $\{ \text{wp}("z = x + 1;", x \neq 0.5 \wedge z > 0) \} = \{ x \neq 0.5 \wedge z > 0 \}$
 $z = x + 1;$
 $\{ x \neq 0.5 \wedge z > 0 \}$

Problem 6 (10 pts., 1 pt. each condition, 1 pt. sufficient/insufficient): Verifying Correctness

For each block of code, fill in all the conditions, then use them to state whether the precondition is sufficient to guarantee the postcondition. If the precondition is insufficient, explain why.

Hint: Use backward reasoning to find the weakest precondition that guarantees the postcondition and see if the given precondition is strong enough to guarantee the postcondition. In other words, is the given precondition not stronger than the weakest precondition?

Copy all code to your answer file and fill in the blanks. Answers must be expressed in the format `wp("code",  $\square$ condition)  $\square$   $\square$ precondition`. Show all work. Assume all referenced variables are defined as integers.

```
(1) { x < 2 }  
    { z < 1 }  
    z = x - z;  
    { w < 1 }  
    w = x - 1;  
    { z < 0 }  
    z = w - 1;  
    { z > 1 }
```

Sufficient or Insufficient: { Insufficient because all variables are defined by a maximum value, not a minimum value (or even a correct one at that) like the post condition states }

```
(2) { x = y  $\wedge$  y > 0  $\vee$  y  $\neq$  x  $\wedge$  x  $\geq$  0 }  
    { x = y  $\wedge$  y > 0  $\vee$  y  $\neq$  x  $\wedge$  x  $\geq$  0  $\wedge$  x < y }  
    if (x < y) {  
        { x < y  $\wedge$  x  $\geq$  0  $\wedge$  x  $\geq$  0 }  
        x--;  
        { x < y  $\wedge$  x  $\geq$  0  $\wedge$  x  $\geq$  0 }  
    } else {  
        { x = y / 2  $\wedge$  y > 0 }  
        x = y / 2;  
        { x = y / 2  $\wedge$  y > 0 }  
    }  
    { x < y  $\wedge$  x  $\geq$  0 }
```

Sufficient or Insufficient: { Insufficient because in the event that x = y and y > 0, we directly conflict with the postcondition. }